

Task 1

Functional Requirements:

- 1 Students and staff can register a vehicle by entering their ID and vehicle details.
- 2 Users can submit a parking permit request for a specific zone.
- 3 Users can check the availability of parking zones in real time.
- 4 Security staff can verify a permit's validity and log parking violations.

Non-functional requirements:

- 1 The system must load permit status within 3 seconds under normal network conditions.
- 2 All users data and permit records must be stored securely with access restricted by role.

Task 4

Defect Analysis

Defect: "A valid parking permit is shown as expired during security verification"

Item	Student Answer
Defect Category	Logic/Data Error
Severity	High
Priority	High
Possible Root Cause	Permit expiry date comparison uses wrong date format or timezone mismatch
Regression Test	Verify all valid permits display correct status after applying the fix